

Assignment 12

Module -13 Windows Server Security and Maintenance

1. Explain the process of installing and configuring Hyper-V virtualization in Windows Server 2016.

❖ **What is Hyper-V**

- Hyper-v is a virtualization technology in Windows Server that allows multiple virtual machines (VMs) to run on one physical server.

❖ **Why Hyper-V is Important**

- It saves hardware cost.
- Multiple operating systems can run on one server.
- Testing and development become easier.
- Better resource utilization.

❖ **Steps to Install Hyper-V**

Step 1: Open Server Manager

- Open Server Manager
- Click:
 - Add Roles and Features

Step 2: Select Installation Type

- Choose:
 - Role-based or feature-based installation

Step 3: Select Server

- Choose your server from the server pool.

Step 4: Select Hyper-V

- Tick:
 - Hyper-V

Step 5: Configure Virtual Switch

Choose the network adapter for virtual machines.

Step 6: Finish Installation

- Click Install
- Restart the server if required.

❖ **How to Configure Hyper-V**

Step 1: Open Hyper-V Manager

Go to:

- Tools → Hyper-V Manager

Step 2: Create Virtual Machine

- Click:
 - New → Virtual Machine

Step 3: **Configure VM**

Enter:

- VM Name
- RAM size
- Storage size
- Network connection
- ISO installation file

Step 4: Start VM

- Click:
 - Start

➤ Now the virtual machine works like a real computer inside the server.

2. How do you monitor server performance and manage event logs in Windows Server?

❖ **Why Monitoring is Important**

- Detect problems early
- Improve performance
- Prevent server crashes
- Analyze hardware and software issues

❖ **Steps to Monitor Performance**

Step 1: Open

- Run → perfmon

Step 2: Select

- Performance Monitor

Step 3: Add Counters such as:

- CPU usage
- Memory usage
- Disk activity
- Network usage

❖ **Why Counters are Important**

➤ They help identify bottlenecks.

❖ **What is Event Viewer?**

- It records system events such as:
- Errors
 - Warnings
 - Login attempts
 - Hardware failures

❖ Steps to Open Event Viewer

- Open:
 - **Server Manager → Tools → Event Viewer**

Important Logs

1. Application Log
2. Security Log
3. System Log

❖ Why Event Logs are Useful

They help troubleshoot:

- Startup failures
- Network problems
- Service crashes
- Login issues

3. Describe the different types of storage options available in Windows Server.

❖ Types of Storage Options

1. Direct Attached Storage (DAS)

- Storage directly connected to the server.
 - Example:
 - Hard Disk
 - SSD
 - **Advantages**
 - Simple
 - Low cost

2. Network Attached Storage (NAS)

- Network Attached Storage provides file storage over a network.
 - **Advantages**
 - Easy file sharing
 - Centralized storage

3. Storage Area Network (SAN)

- Storage Area Network is a high-speed network for block-level storage.
 - **Advantages**
 - High performance
 - Large storage capacity
 - Used in:
 - Datacenters
 - Enterprise environments

4. RAID Storage

- RAID combines multiple disks for:
 - Performance
 - Redundancy
 - Fault tolerance

5. Cloud Storage

- Online storage provided through the internet.
 - Example:
 - Microsoft cloud services

4. What is the role of File Server in Windows Server, and how do you configure it?

❖ What is a File Server?

- File Server stores and shares files across a network.
 - Users can:
 - Upload files
 - Download files
 - Share folders

❖ Why File Server is Important

It provides:

- Centralized file management
- Backup support
- Secure file sharing
- Access control

❖ Steps to Configure File Server

Step 1: Install File Server Role

- Open Server Manager
- Select:
 - Add Roles and Features
- Install:
 - File and Storage Services

Step 2: Create Shared Folder

- Create a folder
- Example:
D:\SharedData

Step 3: Share Folder

- Right-click folder
- Select:
 - Properties → Sharing → Advanced Sharing

Enable:

- Share this folder

Step 4: **Configure Permissions**

Set:

- Read permission
- Write permission
- Full control

➤ Users on the network can now access shared files.

5. Explain the process of implementing and managing Distributed File System (DFS) in Windows Server 2016.

❖ **What is DFS**

➤ Distributed File System allows administrators to organize shared folders from different servers into one virtual location.

❖ **Why DFS is Important**

DFS provides:

- Centralized file access
- File replication
- High availability
- Load balancing

❖ **Types of DFS**

1. DFS Namespace

➤ Creates a virtual shared folder structure.

2. DFS Replication

➤ Copies files automatically between servers.

❖ **Steps to Configure DFS**

Step 1: Install DFS Role

- Open Server Manager
- Install:
 - DFS Namespaces
 - DFS Replication

Step 2: Open DFS Management

Go to:

- Tools → DFS Management

Step 3: **Create Namespace**

- Right-click:
 - Namespaces
- Select:
 - New Namespace

Step 4: **Add Shared Folders**

Add folders from different servers.

Step 5: **Configure Replication**

- Create replication group
- Select servers
- Configure schedule and bandwidth
 - Files stay synchronized between servers automatically.

6. Discuss the built-in backup and recovery options available in Windows Server 2016 or 2019.

❖ **What is Backup?**

- Backup means creating copies of important data.

❖ **Why Backup is Important**

Backup protects data from:

- Hardware failure
- Virus attacks
- Accidental deletion
- System crashes

❖ **Built-in Backup Options**

1. Windows Server Backup

Windows Server Backup is the built-in backup tool.

It supports:

- Full server backup
- System state backup
- File backup

2. System State Backup

Backs up:

- Registry
- Active Directory
- Boot files

3. Bare Metal Recovery

Restores the complete server.

4. Volume Shadow Copy

Creates snapshots of files for quick recovery.

❖ Recovery Options

Recovery can restore:

- Entire server
- Files and folders
- System state
- Volumes

7. How do you configure Windows Server Backup to back up critical data?

❖ Steps to Configure Backup

Step 1: Install Backup Feature

- Open Server Manager
- Add feature:
 - Windows Server Backup

Step 2: Open Backup Tool

Go to:

- Tools → Windows Server Backup

Step 3: Start Backup Wizard

Select:

- Backup Schedule

Step 4: Choose Backup Type

Options:

- Full Server
- Custom Backup

Step 5: Select Backup Destination

Example:

- External drive
- Another volume
- Network location

Step 6: Configure Schedule

Example:

- Daily backup at 10 PM
 - Critical data is automatically backed up regularly.

8. Explain the steps for restoring files and folders using Windows Server Backup.

❖ Steps to Restore Data

Step 1:**Open**

- Windows Server Backup

Step 2:**Select**

- Local Backup

Step 3:**Click**

- Recover

Step 4:**Choose Backup Location**

Select:

- Local drive
or
- Network backup

Step 5:**Choose Recovery Type**

Select:

- Files and folders

Step 6:**Select Files to Restore**

Choose required files or folders.

Step 7:**Choose Restore Location**

- Original location
or
- Alternate location
 - The selected files are restored successfully.

9. What are some common troubleshooting techniques for Windows Server startup issues?

❖ **Common Startup Problems**

- Corrupt boot files
- Driver failur
- Missing system files
- Hardware issues

❖ **Troubleshooting Techniques**

1. Safe Mode

Starts Windows with minimal drivers.

Used to:

- Remove faulty drivers
- Fix software issues

2. Startup Repair

Automatically repairs boot problems.

3. Check Event Logs

Use:

Event Viewer

to identify errors.

4. Use System Restore

Restores previous working settings.

5. Bootrec Commands

Commands:

- bootrec /fixmbr
 - bootrec /fixboot
 - bootrec /rebuildbcd
- Repair boot configuration.

6. Hardware Check

Check:

- RAM
- Hard disk
- Power supply

10. How do you troubleshoot network connectivity problems in Windows Server?

❖ Common Causes

- Wrong IP configuration
- DNS issues
- Cable failure
- Firewall blocking

❖ Troubleshooting Steps

1) Check IP Configuration

Use command:

- ipconfig /all

2) test connectivity

Use command:

- ping

3. Check DNS

Use command:

- **nslookup google.com**

➤ These steps help identify and solve network problems quickly.

11. Discuss common Active Directory-related issues and their troubleshooting steps.

❖ **What is Active Directory?**

Active Directory manages:

- Users
- Computers
- Permissions
- Authentication

❖ **Common Active Directory Problems**

1. Replication Failure

Domain controllers fail to synchronize.

- Solution

Use:

repadmin /replsummary

2. DNS Problems

AD depends heavily on DNS.

- Solution

Check:

- DNS records
- Forwarders
- Zone settings

3. Login Failure

Users cannot log in.

- Solution
- Check account lockout
- Reset password
- Verify network connection

4. Time Synchronization Issues

Kerberos authentication requires correct time.

- Solution

Synchronize system clocks.

5. Group Policy Not Applying

- Solution

Use:

gpupdate /force

12. Explain how to troubleshoot performance problems on Windows Server 2016 or 2019.

❖ Common Performance Problems

- High CPU usage
- Low memory
- Disk bottleneck
- Network congestion

❖ Troubleshooting Methods

1. Use Task Manager

Check:

- CPU
- RAM
- Processes

2. Use Performance Monitor

Performance Monitor tracks:

- Processor usage
- Disk speed
- Memory usage

3. Check Event Logs

Use:

Event Viewer

to identify warnings and errors.

4. Stop Unnecessary Services

Disable unwanted applications and services.

5. Upgrade Hardware

Upgrade:

- RAM
- SSD
- CPU

if resources are insufficient.

6. Check Malware

Viruses can slow down the server.

Use antivirus scanning